GeoAI-Act: GeoAI-based Activity Inference from GPS Trajectories

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Abstract

Mobile device location data provide critical resources for understanding human mobility, yet detecting stays and inferring activities from noisy and irregular GPS trajectories remains challenging. Existing stay detection algorithms often rely on rule-based or clustering methods with manually specified spatiotemporal thresholds, making them sensitive to varying sampling frequency and data noise. Moreover, most methods remain semantically limited, as they can identify where and when a user stops but provide little information about what the stop represents as a human activity. To address these limitations, we propose GeoAI-Act, a two-stage GeoAI framework that converts raw GPS trajectories into semantically labeled stay episodes. The framework integrates GeoAI-Act-Detect, a geographic-context Transformer for point-level stay detection, with GeoAI-Act-Purpose, a Llama-3.1 large language model (LLM) fine-tuned with QLoRA for five-class stay purpose inference. On the Boston-Walks dataset, GeoAI-Act-Detect outperforms rule-based, clustering-based, and supervised baselines, achieving point- and segment-level F1 scores of 0.894 and 0.813, while GeoAI-Act-Purpose achieves 0.808 accuracy and 0.749 Macro-F1, surpassing heuristic, classical machine learning, and prompting-only baselines. External validation on human-validated California Unacast data demonstrates the transferability of both components across regions and data sources, reaching point- and segment-level F1 scores of 0.964 and 0.863 for stay detection and 0.779 accuracy for purpose inference, while feature-group ablations show that removing geographic context causes the largest performance declines in both tasks. These findings demonstrate that sequence modeling, geographic context, and fine-tuned LLMs can jointly support accurate, interpretable, and transferable activity inference from GPS data.

CCS Concepts

• **Information systems** → **Geographic information systems**; **Spatial-temporal systems**; • **Computing methodologies** → **Machine learning approaches**.

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Keywords

GeoAI, Activity detection, Mobile device location data, Large language models, Human mobility

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1 Introduction

Mobile device location data have become a critical source for understanding human mobility [28]. They support a wide range of applications in transportation analysis [32], urban computing [25], public health [24], and disaster mobility research [13]. However, raw GPS trajectories are difficult to use directly because they often contain millions of points, irregular sampling intervals, location noise, and device-level gaps [33]. Many downstream applications require a higher-level unit of analysis that can describe where people stayed, how long they stayed, and what those stays likely represented. Therefore, a key preprocessing step is to identify stay points or stay activities from raw trajectories [30]. A stay point is commonly defined as a location where a device remains within a certain spatial range for longer than a specified time threshold [7]. In practice, stay points help convert noisy GPS points into meaningful activity episodes, which supports trip extraction, home and work location inference, and travel behavior analysis. Without reliable stay detection, errors can propagate through downstream analyses, leading to inaccurate activity identification, mobility metrics, and behavioral inferences that may ultimately affect transportation planning and policy decisions.

Existing stay detection algorithms often rely on predefined spatial and temporal rules [12, 31]. A common approach is to use a distance threshold and a duration threshold to determine whether a sequence of GPS points forms a stay [13, 30]. Other methods use clustering algorithms, such as DBSCAN [7], ST-DBSCAN [3], or toolkit-based approaches such as scikit-mobility [19]. Grid-based methods, such as H3-based approaches [21], summarize local movement patterns by grouping points into spatial cells. These methods are useful and easy to implement, but they still face several limitations. First, manually specified thresholds may not transfer well across users, sampling frequencies, data sources, and geographic contexts. Second, real-world mobile GPS data are often sparse, noisy, and irregularly sampled, which can split true stays into fragmented

segments or create false stops from noisy observations. Third, most existing methods remain semantically limited. They can identify where and when a user stops, but provide little information about what the stop represents as a human activity. This limits their usefulness for activity-based mobility analysis, where the goal is not only to detect stops but also to distinguish meaningful destinations from incidental pauses or movement-related delays.

To address these limitations, we propose GeoAI-Act, a two-stage GeoAI framework that converts raw GPS trajectories into semantically labeled stay episodes. In the first stage, GeoAI-Act-Detect encodes point-level kinematic, temporal, data-quality, and geographic-context features with a Transformer and estimates the stay probability of each GPS point, from which stay segments are constructed. We adopt self-attention because the evidence distinguishing a stay from a brief pause or positioning noise is dispersed over long, irregularly sampled spans of a trajectory and must be aggregated adaptively. In the second stage, GeoAI-Act-Purpose represents each detected stay segment as structured temporal, POI, user-history, and geographic context, and infers its activity purpose with a fine-tuned large language model (LLM). We adopt an LLM because purpose inference relies on commonsense knowledge about places and daily routines that is not contained in the trajectory data itself.

We evaluate GeoAI-Act on the Boston-Walks dataset, comparing stay detection with rule-based, grid-based, clustering-based, and supervised learning baselines at the point, segment, and user-day levels, and comparing purpose inference with majority, rule-based, classical machine learning, and prompting-only LLM baselines, with GeoAI-Act achieving the best overall performance on both tasks. Both components are further externally validated on human-validated California Unacast data to examine transferability across geographic regions and data sources.

This paper makes three main contributions. First, we propose GeoAI-Act, a two-stage spatial AI framework that transforms GPS trajectories into semantically labeled stay episodes. Second, we design and evaluate a Transformer-based stay detector that combines trajectory dynamics, temporal structure, data quality indicators, and geographic context, and assess its performance at the point, segment, and user-day levels as well as on an external validation dataset. Third, we develop an LLM-based stay purpose inference module and show that fine-tuning improves purpose classification over rule-based, machine learning, and prompting-only baselines.

2 Related Work

2.1 Stay Detection from GPS Trajectories

Mobile device location traces are sequences of time-stamped GPS points that capture movement but do not directly describe activities. A common first step in mobility analysis is therefore to separate periods of movement from periods of stay, where a stay point is defined as a location at which a user remains within a specified spatial range for longer than a given time threshold [31]. This step transforms raw trajectories into activity-relevant units that support trip extraction, home and work location inference, activity pattern analysis, and disaster mobility studies [5, 11, 20, 27, 29]. Because these units feed later analyses, inaccurate stay detection propagates into biased behavioral inferences and less reliable planning decisions.

Early methods relied on spatial and temporal rules, identifying a stay when consecutive points remain within a distance threshold for longer than a duration threshold [12, 31], or when velocity falls below a fixed value. Such rules are simple and transparent, but they perform best when sampling intervals are regular and when the selected thresholds match the study context, conditions that are not always met in mobile device data. Clustering-based methods form a second line of work. DBSCAN identifies clusters of arbitrary shape and treats isolated points as noise [7], and ST-DBSCAN extends this idea to spatial and temporal proximity [3]. Related approaches use hierarchical clustering, K-means, density variants, or grid-based representations such as H3 [21], while toolkits such as scikit-mobility standardize common mobility measures [19].

Prior work has also shown that stays are not always clean geometric objects. [4] proposed a time-aware density-based method to extract stay regions with uncertain boundaries from low-sampling-frequency trajectories, and [1] developed a system for stop identification and stop-area analysis. Recent work on trajectory splitting similarly shows that simple rules are often insufficient for reliable trip extraction [17]. Despite this progress, most methods depend on manually selected thresholds or clustering parameters, which makes them sensitive to sampling rate, user behavior, and local urban form, and they characterize stops geometrically rather than behaviorally. We follow this line of research but take a learning-based approach, framing stay detection as a point-level sequence labeling task in which stay probabilities are learned from trajectory features and geographic context.

2.2 Semantic Inference of Stay Purpose

Stay detection identifies where and when a person stops, but many applications also need to know what the stop represents. Movement patterns alone rarely carry this information: a long stay may indicate home, work, or school, while a short stay may reflect shopping, errands, dining, or a transfer. Trip purpose inference has been studied extensively in GPS-based travel surveys, typically by detecting trip ends, selecting contextual features, and applying inference models [18]. Common features include time of day, day of week, stay duration, nearby POIs, land use, and historical travel patterns. [16] combined GPS trajectories, POIs, and geo-tagged social media data to infer travel purposes, other studies imputed trip purposes using machine learning models [8], and related work shows that land use and POI information help explain why a trip is made rather than only where it goes [10].

Geographic context is useful, but it does not resolve ambiguity. POI labels do not map cleanly onto human activities: a restaurant may indicate a work lunch, a social visit, or a family errand, and a residential location may indicate home, caregiving, or a visit. User history reduces this ambiguity only when combined with time, duration, and local context. In GeoAI-Act, geographic context serves both stages. It helps the detector separate activity locations from movement-related pauses, and it provides the semantic cues used for purpose inference.

Language models have recently been applied to mobility and trajectory analysis. [2] used large language models to synthesize and assess urban mobility patterns from travel survey settings, [26] studied prompt mining for language-model-based mobility flow

forecasting, and [14] treated trajectories as ordered sequences for anomaly detection. These studies show that language models can represent mobility context flexibly, but they do not address the semantic interpretation of individual stays from structured stay-level context. GeoAI-Act addresses this gap. Each stay segment is represented with temporal, OSM and POI, user-history, and neighboring trajectory context, and we compare zero-shot prompting, multi-shot prompting, and QLoRA fine-tuning of Llama-3.1 with rule-based and machine learning baselines.

3 Methodology

3.1 Problem Definition

We study two connected tasks: detecting stay segments from raw GPS trajectories, and inferring the activity purpose of each detected segment. Appendix A summarizes the notation used below.

Let $\mathcal{U}$ denote the set of users. For each user $u \in \mathcal{U}$, we observe a time-ordered GPS trajectory $\mathcal{T}^u = \{(\ell_i^u, t_i^u)\}_{i=1}^{n_u}$, where $\ell_i^u = (lat_i^u, lon_i^u)$ is the location of point i, t_i^u is its timestamp with $t_i^u < t_{i+1}^u$, and n_u is the number of observations. For each point we construct the feature vector

$$\mathbf{x}_i^u = [\mathbf{k}_i^u, \mathbf{w}_i^u, \mathbf{r}_i^u, \mathbf{q}_i^u, \mathbf{o}_i^u], \tag{1}$$

whose five groups denote kinematic, window-based trajectory, temporal, data-quality and sampling-gap, and OSM-derived geographic-context features. The trajectory-level feature sequence is $\mathbf{X}^u = [\mathbf{x}_1^u, \ldots, \mathbf{x}_{n_u}^u]$.

Problem 1: Stay Point Detection. Each GPS point carries a binary label $y_i^u \in \{0, 1\}$, where $y_i^u = 1$ indicates that the point belongs to a stay activity and $y_i^u = 0$ indicates movement. The stay detector estimates

$$\hat{p}_i^u = f_\theta(\mathbf{X}^u)_i = P(y_i^u = 1 \mid \mathbf{X}^u), \tag{2}$$

where f_θ denotes the detection model. Point-level probabilities are converted into binary labels using a threshold δ selected on the validation set, so that $\hat{y}_i^u = 1$ if $\hat{p}_i^u \geq \delta$ and $\hat{y}_i^u = 0$ otherwise.

Problem 2: Stay Segment Construction. A predicted stay segment $\hat{s}_j^u = \{i : a_j \leq i \leq b_j,\ \hat{y}_i^u = 1\}$ is a consecutive run of predicted stay points, where a_j and b_j are its start and end indices. Two adjacent stay points are kept in the same segment only if their time gap is smaller than a maximum gap threshold γ, and a segment is retained only if its duration satisfies $t_{b_j}^u - t_{a_j}^u \geq \tau$. This yields the predicted set $\hat{\mathcal{S}}^u = \{\hat{s}_1^u, \ldots, \hat{s}_{m_u}^u\}$. Ground-truth segments $\mathcal{S}^u = \{s_1^u, \ldots, s_{r_u}^u\}$ are constructed from the labeled stay points in the same way. A predicted segment is considered matched to a ground-truth segment when their temporal intersection over union reaches a threshold η:

$$IoU(s, \hat{s}) = \frac{\left|[t_s^{start}, t_s^{end}] \cap [t_{\hat{s}}^{start}, t_{\hat{s}}^{end}]\right|}{\left|[t_s^{start}, t_s^{end}] \cup [t_{\hat{s}}^{start}, t_{\hat{s}}^{end}]\right|} \geq \eta. \tag{3}$$

Problem 3: Stay Purpose Inference. For each stay segment s_j^u we construct a segment-level context vector

$$\mathbf{z}_j^u = [\mathbf{a}_j^u, \mathbf{b}_j^u, \mathbf{h}_j^u, \mathbf{m}_j^u], \tag{4}$$

containing temporal context, geographic and POI context, user-history context, and neighboring trajectory context, respectively. The purpose label c_j^u takes one of five values in the label space C: Home, Work/Study, Personal Needs/Errands, Recreation/Social, and Others. The inference model predicts

$$\hat{c}_j^u = \arg\max_{c \in C} g_\phi(c \mid \mathbf{z}_j^u), \tag{5}$$

where g_ϕ denotes the Llama-3.1-based purpose inference model. Together, f_θ and g_ϕ convert raw GPS trajectories into semantically labeled stay segments.

3.2 Overview of Model Framework

Figure 1 presents the GeoAI-Act framework. Labeled GPS trajectories are sorted by user and time and split by user into training, validation, and test sets, so that no user appears in more than one split. Point-level features are then constructed from the five groups defined above, characterizing both movement patterns and the surrounding built environment. Stage 1 (Section 3.3) applies the geographic-context Transformer to the point-level feature sequence and converts the resulting stay probabilities into stay segments. Stage 2 (Section 3.4) enriches each detected segment with stay-level context and assigns an activity purpose. The final output is a set of semantic stay episodes, each pairing a detected stay segment with an inferred purpose. The framework is evaluated against stay detection and purpose inference baselines at the point, segment, and user-day levels, together with ablation analysis, error analysis, and external validation on California Unacast data.

3.3 Transformer-Based Stay Detection

GeoAI-Act implements f_θ with a Transformer encoder [23], which takes the user-level feature sequence $\mathbf{X}^u$ as input. Since stay detection is conducted offline, both past and future points are available as context. The Transformer matches the structure of this task in three respects. First, a stationary observation may reflect a stay, a traffic delay, or positioning noise, and disambiguation requires evidence dispersed over minutes to hours of surrounding trajectory; self-attention aggregates this evidence through a content-dependent, point-specific context window rather than fixed distance and duration criteria. Second, attention connects any two points in a single step, so the coherence of long stay segments does not rely on information propagating through hundreds of intermediate points, as in recurrent models. Third, attention assumes no uniform sampling: inter-point time gaps enter the model as explicit features, matching the irregular sampling of mobile device data.

The encoder maps $\mathbf{X}^u$ into contextual point representations, and a linear classification layer applied to each representation estimates $\hat{p}_i^u$. The model is trained with cross-entropy loss over labeled GPS points:

$$\mathcal{L}_{stay} = -\frac{1}{\sum_{u \in \mathcal{U}} n_u} \sum_{u \in \mathcal{U}} \sum_{i=1}^{n_u} \log P(y_i^u \mid \mathbf{X}^u). \tag{6}$$

Predicted probabilities are then thresholded at δ, consecutive stay points are merged, segments are split where the time gap exceeds γ, and segments shorter than τ are removed, producing the final predicted stay segments $\hat{\mathcal{S}}^u$.

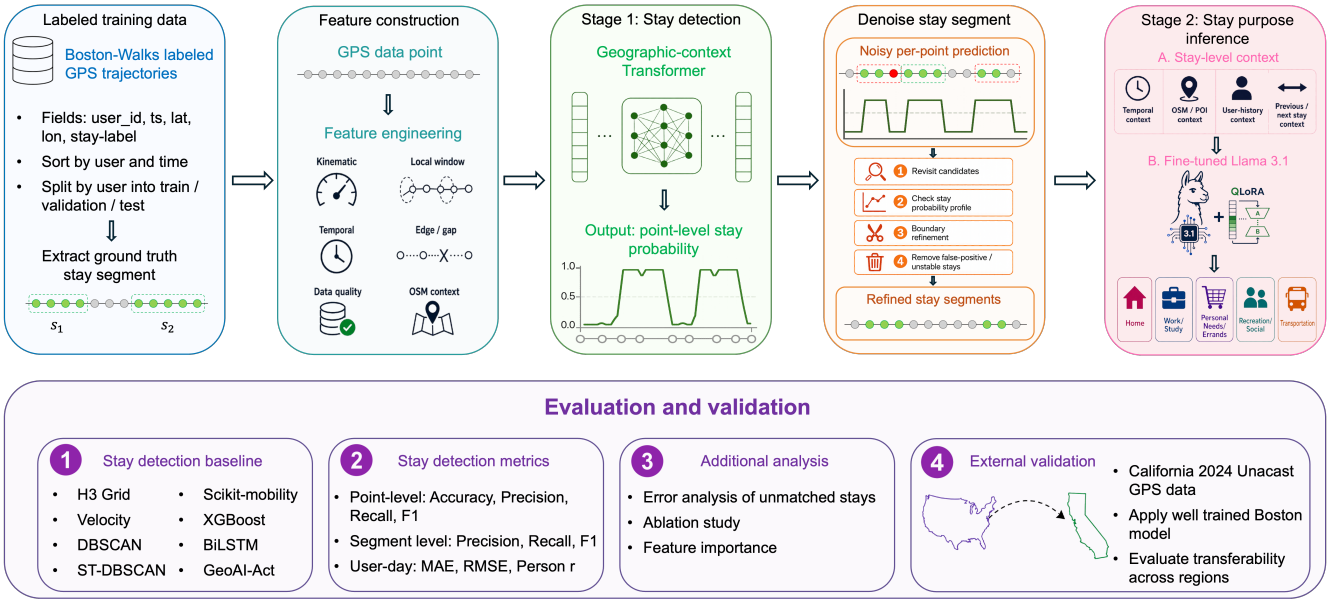


Figure 1: Overview of the GeoAI-Act framework.

3.4 Llama-Based Stay Purpose Inference

GeoAI-Act implements g_ϕ with Llama-3.1 [9]. Unlike stay boundaries, which are determined by information contained in the trajectory, stay meanings depend largely on knowledge external to the data. A pretrained LLM encodes such commonsense priors and represents POI semantics in an open vocabulary, generalizing to POI types that are rare or absent in the training data, whereas conventional classifiers treat POI categories as unordered indices. Fine-tuning then adapts these priors to the label taxonomy and regional context of the study data. For inference, the context vector $\mathbf{z}_j^u$ is converted into a structured prompt, and the model outputs one activity purpose from C.

We fine-tune Llama-3.1 using QLoRA [6]. The base model is quantized and frozen, and only low-rank adapter parameters are updated:

$$\mathbf{W}' = \mathbf{W} + \frac{\alpha}{r}\mathbf{BA}, \tag{7}$$

where $\mathbf{W}$ is a frozen weight matrix, $\mathbf{A}$ and $\mathbf{B}$ are trainable low-rank matrices, r is the adapter rank, and α is a scaling factor. The model is trained to generate the ground-truth purpose label c_j^u from the structured stay context.

3.5 External Validation

We developed two task-specific workflows to evaluate the transferability of GeoAI-Act to vendor-based GPS data from California (see Section 4.2). As illustrated in Figure 2, separate human-validated reference labels were established for GeoAI-Act-Detect and GeoAI-Act-Purpose, which allows the two components to be evaluated independently and distinguishes detection errors from semantic errors.

For stay detection, rule-based methods were first applied to the quality-gated trajectories to generate candidate stay segments. Two annotators independently reviewed the complete trajectories and

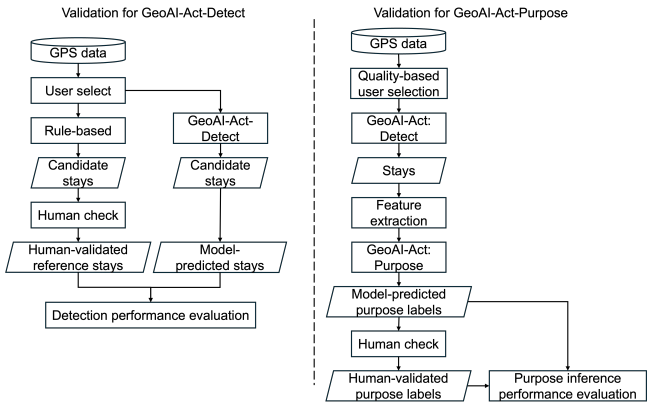


Figure 2: External validation workflow for the two-stage GeoAI-Act framework.

candidate stays using a map-based interface, correcting inaccurate boundaries, removing false stays, and identifying missed stays. Disagreements were resolved to establish the final reference labels, and agreement between the initial candidates and these labels reached a point-level Cohen's κ of 0.976 and a segment-level F1 score of 0.786. GeoAI-Act-Detect was then applied independently to the same trajectories, and its predictions were compared with the reference labels at the point, segment, and user-day levels.

For stay purpose inference, GeoAI-Act-Detect first identified stay segments from the same trajectories, and temporal, geographic, user-history, and neighboring-trajectory features were extracted for each stay to generate an initial purpose prediction. Two annotators independently reviewed the stay-level context and assigned one of the five categories in C, reaching a Cohen's κ of 0.901, and

disagreements were resolved to establish the final reference labels. GeoAI-Act-Purpose predictions were compared with these labels using accuracy, Macro-F1, Weighted-F1, and class-level F1.

3.6 Evaluation Metrics

We evaluate stay detection at three levels. At the point level, we compare $\hat{y}_i^u$ with y_i^u and report accuracy, precision, recall, and F1. At the segment level, we apply the IoU matching rule defined in Section 3 and report precision, recall, F1, and mean matched IoU. At the user-day level, let C_d and $\hat{C}_d$ denote the true and predicted number of stay segments for user-day d, over D user-days. We report

$$MAE = \frac{1}{D}\sum_{d=1}^{D}\left|\hat{C}_d - C_d\right|,$$
$$RMSE = \sqrt{\frac{1}{D}\sum_{d=1}^{D}\left(\hat{C}_d - C_d\right)^2}, \tag{8}$$
$$\Delta = \frac{1}{D}\sum_{d=1}^{D}\left(\hat{C}_d - C_d\right),$$

where Δ is a signed measure of systematic over- or under-detection, together with the Pearson correlation between $\hat{C}_d$ and C_d.

For stay purpose inference, we compare $\hat{c}_j^u$ with c_j^u and report accuracy, class-level F1, Macro-F1 = $\frac{1}{|C|}\sum_{c\in C} F1_c$, and Weighted-F1 = $\sum_{c\in C}\frac{n_c}{N}F1_c$, where n_c is the number of test samples in category c and N is the total number of test samples.

4 Data

4.1 Boston-Walks Dataset

We use the Boston-Walks dataset as the primary labeled dataset for model training, validation, and internal testing. Boston-Walks is a large-scale mobility dataset collected in the Boston metropolitan area through smartphone-based travel diaries [15]. The full dataset contains GPS trajectories from 990 participants, with a median tracking duration of 23 days, about 235,000 recorded tracks, and 172,000 identified stay events. Compared with commercial mobile location datasets, Boston-Walks provides richer behavioral context and more structured annotations. After preprocessing and eligibility filtering, 973 of the 990 participants remained in the analytic sample. The 973 users were divided into 70% for training and 15% for validation and testing, respectively.

The dataset contains detailed GPS trajectories, timestamps, movement records, and activity-related labels. For this study, we use the point-level labels that identify whether each GPS observation belongs to a stay or movement episode. These labels allow us to train and evaluate stay detection models directly. We also use the labeled stay activities with purpose annotations to construct the stay purpose inference task. We preprocess the data by sorting observations by user and time. To reduce leakage, users are split into training, validation, and testing sets, so that records from the same user do not appear in multiple splits. For stay detection, we construct point-level features from their trajectories. Ground-truth stay segments are then extracted from labeled stay points using the duration and gap rules defined in Section 3.

Boston-Walks serves two roles in this paper. First, it provides the labeled trajectories used to train the Transformer-based stay detector. Second, it provides stay segments with activity labels for training and evaluating the Llama-based purpose inference model. Since Boston-Walks is more structured and better annotated than typical vendor-based GPS data, we also use the California Unacast dataset in Section 4.2 to evaluate whether both GeoAI-Act-Detect and GeoAI-Act-Purpose transfer to a different data source and geographic region.

4.2 California Unacast Dataset

We use the California Unacast dataset to externally evaluate the transferability of GeoAI-Act-Detect and GeoAI-Act-Purpose to vendor-based GPS data from a different geographic region. The statewide dataset covers December 1, 2024, to January 28, 2025, and contains more than 27.5 billion observations from anonymous mobile devices [22]. Compared with Boston-Walks, the California data are sparser, less structured, and do not contain provider-supplied stay or activity-purpose labels.

The raw records include device IDs, coordinates, timestamps, and data-quality indicators. We removed observations flagged as spoofed, grid-snapped, or associated with abnormal device clustering and collapsed duplicate temporal observations. A quality-gated validation sample was selected for December 18–20, 2024, based on ping density, multi-day activity, and temporal coverage. The final sample contains 30 users and 39,629 deduplicated GPS pings.

Because provider-supplied labels were unavailable, two annotators established human-validated reference labels for both stay detection and stay purpose inference following the workflows described in Section 3.5 and Figure 2. These labels support empirical external evaluation of both GeoAI-Act components.

5 Results

This section evaluates the two main components of the GeoAI-Act framework. We refer to the stay detection component as *GeoAI-Act-Detect* and the stay purpose inference component as *GeoAI-Act-Purpose*. The first two subsections report the performance of GeoAI-Act-Detect and GeoAI-Act-Purpose on Boston-Walks. The third subsection evaluates the external transferability of both GeoAI-Act-Detect and GeoAI-Act-Purpose on the California Unacast dataset. The final subsection examines the contribution of major feature groups to both framework components.

5.1 Stay Detection Performance

Table 1 reports the stay detection results on the Boston-Walks test set. The dataset includes 973 users, with an average observation period of 35.3 days and 5.15 stays per user-day. GeoAI-Act-Detect is compared with five clustering / rule-based methods and two supervised learning baselines.

The results show that GeoAI-Act-Detect provides the strongest overall performance across the point-, segment-, and user-day evaluation levels. At the point level, GeoAI-Act-Detect achieves the highest accuracy of 0.916 and the highest F1 score of 0.894. At the segment level, GeoAI-Act-Detect achieves the highest precision, recall, and F1 score, at 0.837, 0.791, and 0.813, respectively. Its mean matched intersection over union (IoU) is 0.919. This shows that

GeoAI-Act-Detect balances segment identification and boundary recovery. At the user-day level, GeoAI-Act-Detect achieves the lowest mean absolute error (MAE) of 0.764 stays per user-day. Its mean signed error is −0.098, indicating a small tendency to underestimate the number of stays. Overall, GeoAI-Act-Detect most consistently recovers both individual stay segments and the number of stays observed within each user-day.

We further examined unmatched true stay segments to characterize the remaining errors. Short or sparsely sampled true stays are the most common error type ($n = 581$). These segments contain limited temporal evidence and few GPS observations, making them difficult to distinguish from brief pauses or transfer activities. The second most common error is absorption into a longer predicted segment ($n = 156$), in which multiple nearby stays are merged. Boundary mismatch accounts for 78 cases, while other errors and low-confidence model predictions account for 53 and 12 cases, respectively. The remaining errors therefore arise primarily from sparse sampling and uncertain segment boundaries rather than from arbitrary point-level misclassification.

5.2 Stay Purpose Inference Results

Table 2 reports the stay purpose inference results on the test set. GeoAI-Act-Purpose, implemented using Llama-3.1 with QLoRA fine-tuning, achieves the strongest overall performance, with an accuracy of 0.808, a Macro-F1 of 0.749, and a Weighted-F1 of 0.807. It outperforms the majority and rule-based baselines, all classical machine learning models, and both zero-shot and multi-shot Llama-3.1 prompting.

Among the non-LLM models, XGBoost performs best, with an accuracy of 0.622, a Macro-F1 of 0.546, and a Weighted-F1 of 0.582. Relative to XGBoost, GeoAI-Act-Purpose improves accuracy by 0.186 and Macro-F1 by 0.203. QLoRA fine-tuning also improves over multi-shot prompting by 0.105 in accuracy, 0.058 in Macro-F1, and 0.113 in Weighted-F1. These results indicate that task-specific fine-tuning helps the model connect temporal, spatial, geographic, and user-history features with activity-purpose semantics.

The class-level results show that Home and Work/Study are the most reliably inferred purposes, with F1 scores of 0.950 and 0.914, respectively. These categories exhibit comparatively stable temporal patterns and repeated user-specific location histories. Others achieves an F1 score of 0.747, suggesting that the model can identify many stays characterized by transfer, waiting, assistance, conflicting evidence, or insufficient contextual information.

Personal Needs/Errands is more difficult, with an F1 score of 0.653, because it combines heterogeneous activities such as eating, shopping, medical visits, and other personal services. Recreation/-Social is the most difficult class, with an F1 score of 0.479. This category contains fewer labeled examples and overlaps with both residential and commercial contexts. For example, a social visit may occur at another person's residence, while a recreational activity may share nearby POIs with dining or personal errands.

5.3 External validation on the California Dataset

Table 3 reports the external stay detection results on the California Unacast dataset. All supervised and fine-tuned models were trained using Boston-Walks and applied to California without retraining or threshold tuning. California labels were used only for final evaluation. GeoAI-Act-Detect achieves the highest point-level accuracy and F1 score, at 0.952 and 0.964, respectively. ST-DBSCAN produces the highest point-level recall of 0.995, while H3Grid produces the highest precision of 0.977. However, GeoAI-Act-Detect provides the strongest balance between precision and recall.

At the segment level, GeoAI-Act-Detect achieves the highest recall of 0.922, F1 score of 0.863, and mean IoU of 0.972. BiLSTM obtains a marginally higher segment precision of 0.815, compared with 0.810 for GeoAI-Act-Detect. At the user-day level, GeoAI-Act-Detect produces the lowest MAE of 1.109, the lowest RMSE of 2.054, the smallest absolute count bias of 0.150, and the highest Pearson correlation of 0.932. The relatively high point- and segment-level performance may partly reflect the quality-gated sample, which contains users with relatively dense observations and broad temporal coverage. However, the user-day MAE increases from 0.764 on Boston-Walks to 1.109 on the California data, indicating that exact daily stay counts remain sensitive to irregular vendor sampling and accumulated segment-splitting or merging errors.

Table 4 reports the external-validation performance of the stay purpose models on the California Unacast dataset. All models were evaluated against the human-validated purpose labels established through independent review by two annotators. GeoAI-Act-Purpose achieves the strongest overall performance, with an accuracy of 0.779, a Macro-F1 of 0.693, and a Weighted-F1 of 0.777. It outperforms the majority and heuristic baselines, the classical machine-learning models, and the zero-shot and multi-shot Llama-3.1 baselines. Relative to its Boston-Walks performance, accuracy decreases by 0.029, Macro-F1 by 0.056, and Weighted-F1 by 0.030, indicating modest performance degradation across datasets. At the class level, Home and Work/Study remain the most accurately inferred categories, with F1 scores of 0.925 and 0.875, respectively. Personal Needs/Errands achieves an F1 score of 0.600, while Recreation/Social remains the most difficult category, with an F1 score of 0.360. Others achieves an F1 score of 0.705. These results suggest that routine activities associated with repeated locations and regular temporal patterns transfer more reliably than heterogeneous activities with greater semantic overlap.

5.4 Ablation Study

Table 5 reports the feature-group ablation results for GeoAI-Act-Detect. One feature group is removed at a time, and the model is retrained using the same data split and evaluation procedure. Removing OSM context produces the largest overall decline. Point-level F1 decreases from 0.894 to 0.792, segment-level F1 decreases from 0.813 to 0.717, and user-day MAE increases from 0.764 to 1.137. Removing kinematic, temporal, or window-based features also reduces performance, but the effects are smaller. These results show that geographic context contributes substantially to distinguishing meaningful stays from trajectory pauses. At the same time, the declines observed after removing the other feature groups indicate that robust stay detection requires both movement dynamics and contextual information.

Table 1: Stay detection performance on the Boston-Walks test set.

	Point-level				Segment-Level (IoU)				User-Day Stay Count			
	Accuracy	Precision	Recall	F1	Precision	Recall	F1	IoU (mean)	MAE	RMSE	Daily MAE difference	Pearson correlation
Cluster / rule-based												
H3Grid	0.843	0.838	0.628	0.718	0.654	0.643	0.648	0.881	1.419	2.373	-0.222	0.830
Velocity	0.483	0.372	0.916	0.530	0.451	0.574	0.505	0.828	2.284	4.438	1.589	0.759
DBSCAN	0.690	0.628	0.058	0.106	0.676	0.070	0.128	0.917	5.265	6.472	-5.236	0.078
ST-DBSCAN	0.780	0.835	0.360	0.503	0.708	0.318	0.439	0.915	3.284	4.325	-3.217	0.631
ScikitMobility	0.786	0.754	0.483	0.589	0.475	0.342	0.397	0.787	2.601	3.937	-1.648	0.693
Learning-based												
XGBoost	0.898	0.815	0.878	0.845	0.815	0.745	0.779	0.888	0.920	1.570	-0.502	0.916
BiLSTM	0.896	0.779	0.902	0.836	0.767	0.725	0.745	0.883	0.834	1.364	-0.325	0.932
GeoAI-Act-Detect	**0.916**	**0.873**	**0.917**	**0.894**	**0.837**	**0.791**	**0.813**	**0.919**	**0.764**	**1.312**	**-0.098**	**0.935**

Table 2: Stay purpose inference performance.

Model	Accuracy	Macro-F1	Weighted-F1	Home F1	Work/Study F1	Personal Needs/Errands F1	Recreation/Social F1	Others F1
rule-based								
Majority	0.311	0.095	0.147	0.474	0	0	0	0
Rule-Based Heuristic	0.365	0.256	0.232	0.431	0.667	0	0.182	0
ML-based								
Logistic Regression	0.351	0.300	0.266	0.545	0.615	0	0.340	0
MLP	0.514	0.451	0.422	0.867	0.684	0.083	0.167	0.455
Random Forest	0.541	0.513	0.545	0.857	0.759	0.537	0	0.414
XGBoost	0.622	0.546	0.582	0.815	0.774	0.596	0	0.545
LLM-based								
Llama-3.1 Zero-Shot	0.662	0.631	0.641	0.897	0.813	0.550	0.250	0.645
Llama-3.1 Multi-Shot	0.703	0.691	0.694	0.929	0.875	0.585	0.400	0.667
Llama-3.1 + QLoRA	**0.808**	**0.749**	**0.807**	**0.950**	**0.914**	**0.653**	**0.479**	**0.747**

Table 3: External validation on the California Unacast dataset for stay detection.

	Point-level				Segment-Level (IoU)				User-Day Stay Count			
	Accuracy	Precision	Recall	F1	Precision	Recall	F1	IoU (mean)	MAE	RMSE	Daily MAE difference	Pearson correlation
Cluster / rule-based												
H3Grid	0.835	0.977	0.768	0.860	0.418	0.684	0.518	0.901	4.967	8.685	4.744	0.505
Velocity	0.646	0.679	0.879	0.766	0.347	0.663	0.456	0.869	7.067	11.102	6.756	0.532
DBSCAN	0.901	0.965	0.881	0.921	0.480	0.566	0.519	0.908	1.522	2.783	1.033	0.910
ST-DBSCAN	0.935	0.914	0.995	0.952	0.646	0.784	0.708	0.918	3.089	4.128	1.333	0.713
ScikitMobility	0.928	0.905	0.955	0.929	0.775	0.835	0.804	0.925	2.456	3.716	-1.589	0.849
Learning-based												
XGBoost	0.930	0.895	0.975	0.933	0.750	0.860	0.801	0.922	1.451	2.672	-0.654	0.895
BiLSTM	0.945	0.925	0.975	0.949	0.815	0.885	0.849	0.935	1.353	2.457	-0.486	0.918
GeoAI-Act-Detect	**0.952**	**0.944**	**0.985**	**0.964**	**0.810**	**0.922**	**0.863**	**0.972**	**1.109**	**2.054**	**-0.150**	**0.932**

Table 6 presents the corresponding ablation results for GeoAI-Act-Purpose. Removing OSM context again causes the largest performance loss. Accuracy decreases from 0.808 to 0.741, Macro-F1 decreases from 0.749 to 0.652, and Weighted-F1 decreases from 0.807 to 0.727. The largest class-level declines occur for Personal Needs/Errands, from 0.653 to 0.495, and Recreation/Social, from 0.479 to 0.295. These categories depend heavily on nearby land-use, building, and POI information. Removing kinematic, temporal, or window-based features results in smaller but consistent reductions in overall performance. The findings indicate that GeoAI-Act-Purpose benefits from combining stable temporal routines and user-history patterns with geographic context. The complete feature set consistently outperforms all evaluated ablated variants.

5.5 Temporal Patterns of Detected Stays and Inferred Purposes

Beyond classification accuracy, we examined whether the inferred purposes reproduce plausible daily activity rhythms. Figures 3–5 present these descriptive checks. They are not formal evaluation metrics. They serve as a behavioral validity check on the semantic outputs of GeoAI-Act.

Table 4: External validation on the California Unacast dataset for stay purpose inference.

Model	Accuracy	Macro-F1	Weighted-F1	Home F1	Work/Study F1	Personal Needs/Errands F1	Recreation/Social F1	Others F1
Rule-based								
Majority	0.439	0.122	0.268	0.610	0.000	0.000	0.000	0.000
Rule-Based Heuristic	0.390	0.286	0.321	0.500	0.580	0.120	0.130	0.100
ML-based								
Logistic Regression	0.410	0.326	0.362	0.560	0.580	0.150	0.220	0.120
MLP	0.515	0.432	0.534	0.810	0.620	0.200	0.150	0.380
Random Forest	0.552	0.460	0.571	0.780	0.680	0.420	0.100	0.320
XGBoost	0.608	0.532	0.637	0.800	0.720	0.520	0.160	0.460
LLM-based								
Llama-3.1 Zero-Shot	0.650	0.586	0.687	0.860	0.760	0.500	0.220	0.590
Llama-3.1 Multi-Shot	0.691	0.639	0.727	0.895	0.820	0.540	0.320	0.620
Llama-3.1 + QLoRA	**0.779**	**0.693**	**0.777**	**0.925**	**0.875**	**0.600**	**0.360**	**0.705**

Table 5: Feature-group ablation results for GeoAI-Act-Detect.

	Point-level				Segment-Level (IoU)				User-Day Stay Count			
	Accuracy	Precision	Recall	F1	Precision	Recall	F1	IoU (mean)	MAE	RMSE	Daily MAE difference	Pearson correlation
GeoAI-Act-Detect	0.916	0.873	0.917	0.894	0.837	0.791	0.813	0.919	0.764	1.312	-0.098	0.935
w/o Kinematic	0.895	0.805	0.886	0.843	0.797	0.727	0.761	0.886	1.027	1.624	0.020	0.910
w/o Temporal	0.899	0.807	0.894	0.848	0.796	0.738	0.766	0.887	0.952	1.481	-0.055	0.925
w/o Window-based	0.894	0.786	0.915	0.845	0.793	0.737	0.764	0.886	0.975	1.553	-0.032	0.916
w/o OSM Context	0.849	0.704	0.904	0.792	0.758	0.681	0.717	0.872	1.137	1.774	0.130	0.893

Table 6: Feature-group ablation results for GeoAI-Act-Purpose.

Model	Accuracy	Macro-F1	Weighted-F1	Home F1	Work/Study F1	Personal Needs/Errands F1	Recreation/Social F1	Others F1
Llama-3.1 + QLoRA	0.808	0.749	0.807	0.950	0.914	0.653	0.479	0.747
w/o Kinematic	0.784	0.719	0.781	0.935	0.885	0.610	0.430	0.735
w/o Temporal	0.789	0.724	0.783	0.930	0.870	0.625	0.440	0.755
w/o Window-based	0.783	0.721	0.780	0.925	0.875	0.620	0.435	0.750
w/o OSM Context	0.741	0.652	0.727	0.920	0.845	0.495	0.295	0.705

Figure 3 shows the share and duration of each inferred purpose. Home accounts for the largest share (32.5%), followed by Recreation/Social (26.8%) and Work/Study (24.6%). Personal Needs/Errands (7.3%) and Other/Unknown (8.8%) are less frequent. The duration patterns match expected activity characteristics. Home stays last the longest, followed by Work/Study. Personal Needs/Errands and Recreation/Social stays show short to medium durations. Other/Unknown stays are the shortest. This ordering indicates that the model distinguishes long routine anchors from brief out-of-home stops.

Figure 4 summarizes hourly and day-of-week variation. Home stays concentrate overnight and in the late evening. Work/Study stays cluster in weekday daytime hours and nearly disappear on weekends. Personal Needs/Errands and Recreation/Social stays spread across afternoon and evening hours and reach their highest shares on weekends. Other/Unknown stays peak in the late afternoon, consistent with transit stops and mode transfers assigned to this class. These patterns indicate that the inference draws on temporal, user-history, and neighboring trajectory context rather than POI matching alone.

Figure 5 compares inferred and ground-truth stay counts per user by hour. The inferred counts track the ground-truth curves on both weekdays and weekends. Both sources show minimal activity in the early morning and a steady rise after mid-morning. Weekday activity peaks in the late afternoon and again in the late evening. Weekend activity concentrates in the afternoon and evening with a flatter profile. These results suggest that GeoAI-Act preserves aggregate temporal structure, even though individual stay boundaries and labels may still contain errors.

6 Discussion

This study developed GeoAI-Act, a two-stage GeoAI framework that converts raw GPS trajectories into semantically labeled stay episodes. The results support four main findings. First, sequence-based modeling achieved the strongest overall balance across point-level classification, stay-segment recovery, and user-day stay counts. Second, fine-tuning Llama-3.1 improved stay purpose inference over rule-based, classical machine-learning, and prompting-only

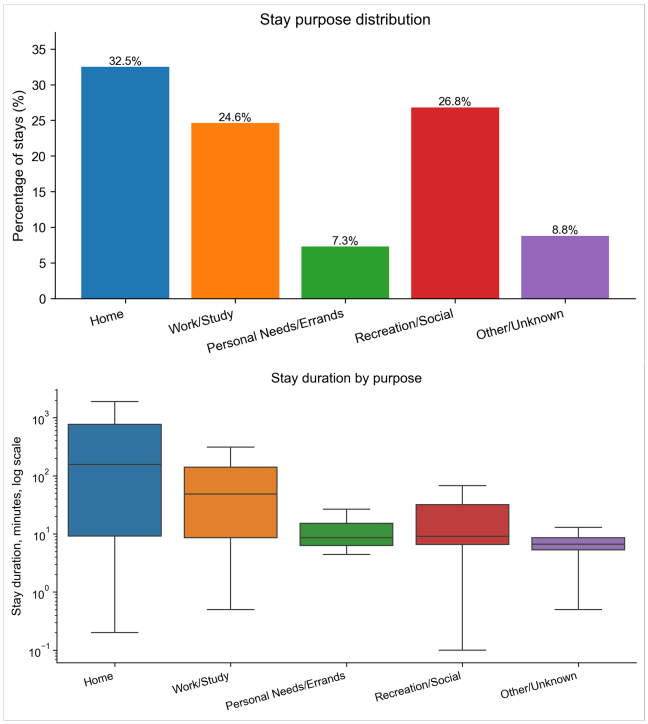


Figure 3: Distribution and duration of inferred stay purposes.

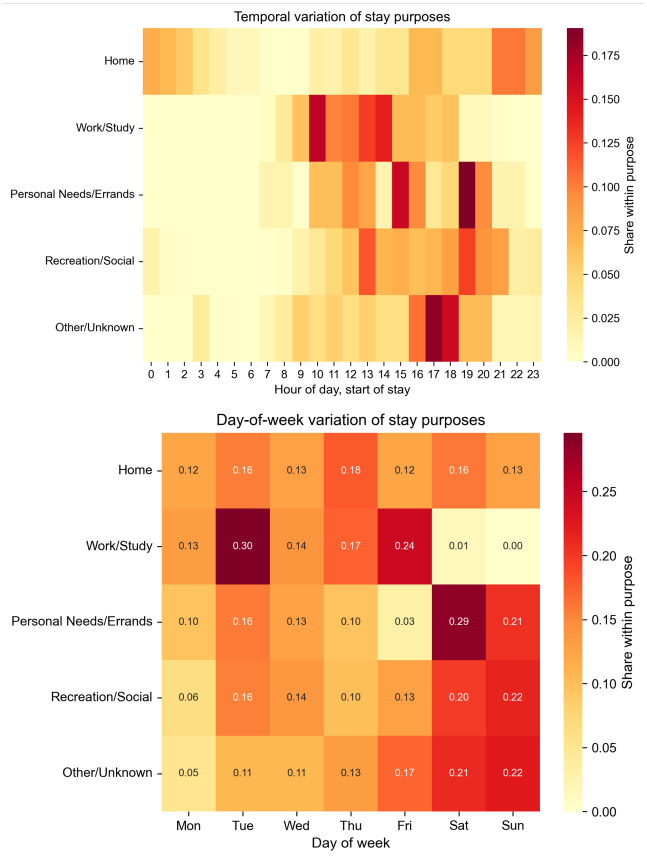


Figure 4: Temporal and day-of-week variation of inferred stay purposes.

approaches. Third, OSM-based geographic context contributed substantially to both components. Fourth, both components retained strong performance on human-validated California vendor data.

On the Boston-Walks test set, GeoAI-Act-Detect achieved a point-level F1 of 0.894, a segment-level F1 of 0.813, and a user-day MAE of 0.764. Its gains over rule-based and clustering baselines reflect the difficulty of selecting fixed geometric or movement thresholds under irregular sampling [1, 4, 17]. The error analysis shows that the remaining errors concentrate in short, sparsely sampled, and boundary-ambiguous stays. Some true stays contained too few observations to be separated from brief pauses, and others were absorbed into longer predicted segments. Sequence modeling uses surrounding context to distinguish sustained activities from noise, but it cannot recover episodes that are weakly represented in the raw data. Reliable stay inference therefore requires both accurate point-level classification and temporally coherent segment construction.

GeoAI-Act-Purpose achieved the strongest purpose inference performance on Boston-Walks, with an accuracy of 0.808, a Macro-F1 of 0.749, and a Weighted-F1 of 0.807. Prompting alone can use some semantic information, but fine-tuning provides a more consistent mapping from structured mobility context to a fixed set of domain-specific activity labels. This result extends recent language-model applications in mobility analysis to the semantic interpretation of individual stay episodes [2, 14, 26].

A shared finding across both components is the value of geographic context. Removing OSM context caused the largest decline among all feature groups in both tasks. Detection point- and segment-level F1 decreased to 0.792 and 0.717, and purpose Macro-F1 decreased to 0.652. Geographic context supports the two stages in complementary ways. During detection, it helps distinguish destination areas from low-speed movement and incidental pauses. During purpose inference, it provides semantic evidence about the functions of nearby buildings, roads, and POIs. The declines from removing other feature groups further indicate that geographic context complements, rather than replaces, trajectory dynamics, temporal routines, and behavioral history [10, 16, 18].

The California external validation indicates that both components transfer from a structured travel-diary dataset to commercial vendor-based GPS data. Against human-validated reference labels, GeoAI-Act-Detect achieved point- and segment-level F1 scores of 0.964 and 0.863. GeoAI-Act-Purpose achieved an accuracy of 0.779 and a Macro-F1 of 0.693, remaining the strongest among the evaluated methods. Some sensitivity to domain shift remains. User-day MAE increased from 0.764 to 1.109, suggesting that small splitting and merging errors accumulate when daily stay counts are derived from irregular vendor trajectories. The modest decreases in purpose metrics may reflect differences in sampling frequency, trajectory completeness, and activity distributions. Because the sample included only 30 quality-gated users over three days, these results

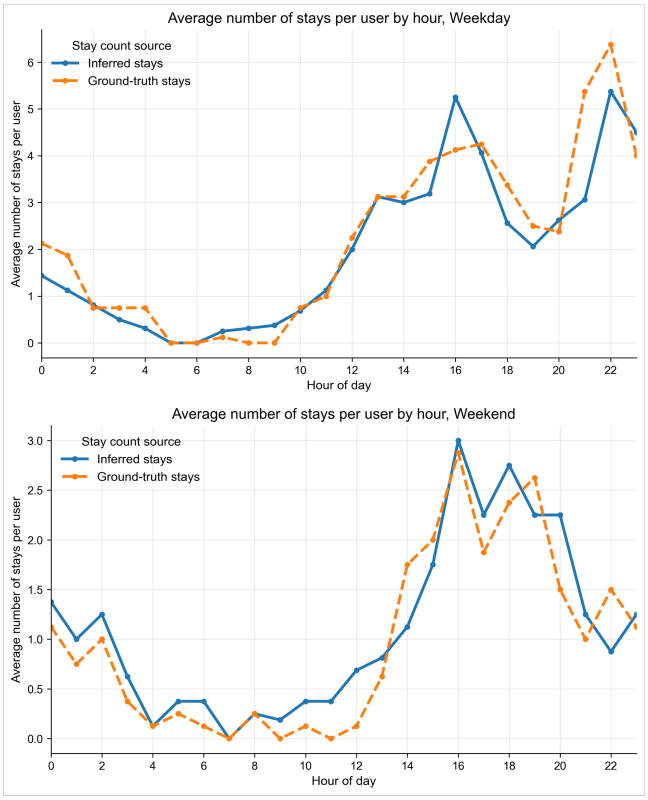


Figure 5: Average number of inferred and ground-truth stays per user by hour on weekdays and weekends.

indicate transferability under the selected data-quality conditions rather than universal generalization [28].

The class-level results clarify where semantic ambiguity remains. Home and Work/Study achieved F1 scores of 0.875 or higher on both datasets, whereas Personal Needs/Errands and Recreation/Social remained at or below 0.653. Home and Work/Study are anchored by repeated locations, longer durations, and regular temporal routines. The weaker categories contain heterogeneous activities that can occur at similar locations and times. A restaurant, commercial area, or residential location can represent several purposes, and geographic categories do not map deterministically to human behavior [16, 18].

This study has several limitations. First, training and internal evaluation relied on Boston-Walks, which represents one metropolitan context and a structured travel-diary collection process. Second, the California validation used a small, quality-gated sample of 30 users over three days. Third, the California reference labels were established through model-assisted annotation, so initial candidate labels may have influenced human review. Fourth, the external purpose evaluation was conducted on stays identified by GeoAI-Act-Detect and therefore does not capture the propagation of detection errors through the full pipeline. Fifth, the five-class taxonomy contains broad and partly overlapping categories. Finally, OSM and POI coverage varies across regions and may introduce geographic data biases.

Future work should validate GeoAI-Act on larger vendor-based datasets covering more regions, sampling frequencies, and data-quality conditions. Annotation studies could use larger samples, additional annotators, and blinded review procedures. Complete pipeline evaluation is needed to quantify how detection errors affect downstream purpose inference. Methodologically, uncertainty estimation could flag sparse stays, ambiguous boundaries, and low-confidence predictions for human review. The taxonomy could be refined into finer-grained purposes such as shopping, dining, healthcare, education, and social visits. Finally, jointly optimized models could allow semantic evidence to support segmentation while improved stay boundaries support purpose inference.

7 Conclusion

This study proposed GeoAI-Act, a two-stage GeoAI framework for detecting stay activities and inferring their semantic purposes from raw GPS trajectories. GeoAI-Act-Detect combines trajectory dynamics, temporal structure, data-quality indicators, and OSM-based geographic context, while GeoAI-Act-Purpose uses a QLoRA-fine-tuned Llama-3.1 model to integrate temporal, geographic, user-history, and neighboring-trajectory context.

External validation using human-validated California Unacast data showed that both components retained strong performance across regions and data sources. GeoAI-Act-Detect achieved point- and segment-level F1 scores of 0.964 and 0.863, while GeoAI-Act-Purpose achieved an accuracy of 0.779, a Macro-F1 of 0.693, and a Weighted-F1 of 0.777. The ablation experiments further showed that OSM-based geographic context contributed substantially to both stay detection and purpose inference. Overall, GeoAI-Act provides a transferable framework for converting sparse and noisy GPS trajectories into semantically meaningful activity episodes.

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A Nomenclature

Table 7 summarizes the notation, parameters, and variables used in the methodology section, along with their corresponding definitions.

B Prompt Template for Stay Purpose Inference

GeoAI-Act uses a structured prompt to convert each detected stay segment into an input for Llama-3.1. The same prompt structure is used for zero-shot prompting, multi-shot prompting, and QLoRA fine-tuning. During QLoRA fine-tuning, the assistant response is the ground-truth purpose label. During inference, the model generates the purpose label.

Prompt Template

System prompt. You are an expert in human mobility analysis. Infer the purpose of a stay segment using temporal, spatial, mobility, POI, and user-history context. Choose exactly one purpose from: Home, Work/Study, Personal Needs/Errands, Recreation/Social, Transportation. Return only valid JSON.

User prompt. The input contains structured stay-level context, including: (1) temporal features such as start hour, end hour, duration, day of week, weekend indicator, and overnight indicator; (2) geographic and POI features such as stay center, nearest POI category, distance to POIs, building indicator, and nearby POI count; (3) user-history features such as night anchor, daytime anchor, repeated visits to the same area, and prior visits at night or on weekdays; and (4) neighboring trajectory features such as travel time from the previous stay, distance from the previous stay, and previous or next POI context.

Output format. `"purpose": "Work/Study"`

Table 7: Notation used in this paper.

Symbol	Definition
$\mathcal{U}$	Set of users
u	A user, where $u \in \mathcal{U}$
$\mathcal{T}^u$	GPS trajectory of user u
n_u	Number of GPS observations for user u
i	Index of a GPS point
j	Index of a stay segment
ℓ_i^u	Location of point i for user u
lat_i^u	Latitude of point i for user u
lon_i^u	Longitude of point i for user u
t_i^u	Timestamp of point i for user u
$\mathbf{x}_i^u$	Feature vector of point i for user u
$\mathbf{X}^u$	Sequence of feature vectors for user u
$\mathbf{k}_i^u$	Kinematic feature vector of point i
$\mathbf{w}_i^u$	Window-based trajectory feature vector of point i
$\mathbf{r}_i^u$	Temporal feature vector of point i
$\mathbf{q}_i^u$	Data quality and sampling-gap feature vector of point i
$\mathbf{o}_i^u$	Geographic context feature vector of point i
y_i^u	Ground-truth binary stay label of point i
$\hat{y}_i^u$	Predicted binary stay label of point i
$\hat{p}_i^u$	Predicted probability that point i is a stay point
f_θ	Transformer-based stay detection model
θ	Parameters of the stay detection model
δ	Probability threshold for converting stay probabilities into labels
τ	Minimum duration threshold for retaining a stay segment
γ	Maximum time gap allowed within a stay segment
$\hat{s}_j^u$	Predicted stay segment j for user u
s_j^u	Ground-truth stay segment j for user u
a_j	Start point index of stay segment j
b_j	End point index of stay segment j
$\hat{\mathcal{S}}^u$	Set of predicted stay segments for user u
$\mathcal{S}^u$	Set of ground-truth stay segments for user u
m_u	Number of predicted stay segments for user u
r_u	Number of ground-truth stay segments for user u
$IoU(s, \hat{s})$	Temporal intersection-over-union between a true and predicted segment
η	IoU threshold for matching predicted and ground-truth stay segments
$\mathbf{z}_j^u$	Segment-level context vector for stay segment j
$\mathbf{a}_j^u$	Temporal context of stay segment j
$\mathbf{b}_j^u$	Geographic and POI context of stay segment j
$\mathbf{h}_j^u$	User-history context of stay segment j
$\mathbf{m}_j^u$	Neighboring trajectory context of stay segment j
c_j^u	Ground-truth purpose label of stay segment j
$\hat{c}_j^u$	Predicted purpose label of stay segment j
$\mathcal{C}$	Set of activity purpose categories
g_ϕ	Llama-based stay purpose inference model
ϕ	Parameters of the stay purpose inference model